AMARTYA BHATTACHARYA

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EDUCATION

Northeastern University **Boston, MA**
Candidate for Bachelor of Science in Computer Engineering / Computer Science *Expected May 2029*
GPA: 3.79 · Dean's List (all semesters)
Relevant Coursework: Algorithms and Data · Program Design and Implementation 2 · Logic and Computation · Embedded Design: Enabling Robotics · Circuits and Signals · Discrete Structures · Differential Equations

SKILLS

Programming Languages: Python, Java, MicroPython, C++, Lean 4, MATLAB
Tools & Frameworks: FastAPI, Anthropic API, Git, Matplotlib, Claude Code, LTspice
Hardware & Instrumentation: Oscilloscopes, waveform generators, op-amp circuit design, I2C interfacing, Raspberry Pi Pico (RP2040)
Languages: Bengali (advanced), Spanish (conversational)

RESEARCH

Material Analysis for the Marimba | *Independent Researcher* | *Plymouth, MN* *Nov. 2024 – Present*
- Investigate alternatives to Honduran rosewood, the primary bar material in marimbas, which is increasingly costly and scarce.
- Evaluate 17 candidate materials by density, specific modulus (E/ϱ), and loss factor, identifying hickory and cherry as top substitutes.
- Measure Young's modulus and loss factor by striking samples suspended by nodes and analyzing their resonant frequency.
- Presented at the Acoustical Society of America's 190th Meeting (Philadelphia, 2026); accepted for publication in Proceedings of Meetings on Acoustics; covered by NPR's *All Things Considered* and NPR's *Short Wave*.

PROJECTS

Wave-Tilter — Tilt-Controlled Waveform Generator | *Personal Project* | *Boston, MA* *Fall 2026*
- Built a tilt-controlled waveform generator on an M5Stack Cardputer ADV (ESP32-S3) with Claude Code.
- Mapped IMU tilt to morph a triangle wave toward square or sine and to glide pitch, with a key-toggled semitone snap.
- Verified outputs on an oscilloscope, then tested an RC low-pass filter (1 kΩ checked with a DMM, 100 µF, 1.6 Hz cutoff).

Solo-Soloing — Jazz Practice Track Generator | *Personal Project* | *Maple Grove, MN* *Summer 2026*
- Built a web app that generates jazz backing tracks (bass, piano, drums), using a FastAPI pipeline in which Claude writes each track as Strudel code, rendered as audio in the browser.
- Designed a 9-check verification suite validating every generated note against music-theory rules for key, form, register, and feel.
- Routed each failed check back to the pipeline stage able to fix it, with a 3-attempt retry loop per LLM stage.

Analog ECG Front-End | *Circuits and Signals* | *Boston, MA* *Spring 2026*
- Designed and built a three-stage analog ECG front-end: an AD627 instrumentation amplifier (gain ≈ 96) feeding active high-pass and low-pass filters on an LT1490 op-amp.
- Set filter cutoffs at 0.40 Hz and 15.9 Hz to reject electrode DC offset and high-frequency noise while keeping the QRS complex clearly identifiable and preventing aliasing at a 150 Hz sampling rate.
- Concentrated all gain in the amplifier stage with unity-gain filters, scaling 1–3 mV cardiac signals up without clipping the supply rails.
- Verified filter response with sine-wave sweeps (1 kHz attenuated to 10–20%) and acquired ECG data in MATLAB.

Space Racing — Embedded Arcade Game | *Cornerstone of Engineering* | *Boston, MA* *Fall 2025*
- Interfaced 3 analog sensors and an I2C LCD/keypad module to a Raspberry Pi Pico in MicroPython, handling sensor thresholding, timing, and interrupt-driven state transitions.

WORK EXPERIENCE

Cindy's Cinnamon Nuts | *Sales Associate* | *Minnesota State Fair, St. Paul, MN* *Summer 2024*
- Produced cinnamon-glazed nuts in batches, timing each roast to keep quality consistent and supply matched to peak demand.

LEADERSHIP & ACTIVITIES

Music for Missions Charity Concert | *Audio Engineer & Coordinator* | *Hopkins, MN* *Jul. 2025, Jul. 2026*
- Ran audio for a ~100-attendee student benefit concert: placed microphones, configured monitor mixes, and mixed live.
- Cued backing-track playback, briefed performers on call times, and arranged the venue, keeping the program running on schedule.

Asian American Students in Action | *Mentor (previously Mentee)* | *Boston, MA* *Sep. 2025 – Present*
- Mentor peers within a cultural community organization, supporting their identity exploration and involvement in student-led initiatives.

NU Aaroh | *Assistant Music Director* | *Boston, MA* *Sep. 2025 – Present*
- Coordinate with the music director team to select repertoire and assign musicians across 27+ roles; perform on dhol and drum set.

NU Sound (Acoustical Society of America Student Chapter) | *Participant* | *Boston, MA* *Sep. 2025 – Present*
- Engage with music technology and acoustics topics alongside peers, complementing independent marimba acoustics research.

INTERESTS

- Cooking East and South Asian food, from Bengali kosha mangsho to Korean galbi
- Music from metal to jazz · Competitive trapshooting · Lifting · Travel (13 countries, 28 U.S. states and territories)